

Block-Based Quantum Image Representation for Resource-Efficient Quantum Image Processing

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Abstract

Quantum Image Processing (QIP) is an area of research that focuses on using the power of quantum computing to represent and analyze images. For QIP to work effectively, there needs to be a Quantum Image Representation (QIR) that can be scaled up to be able to use the available quantum computers. Many models of QIR, such as FRQI, NEQR, and GQIR, already exist; however, most of these models utilize uniform pixel-based encoding for classical images which causes these models to use a lot of qubits, creating deep quantum circuits, making them not practical on the current Noisy Intermediate-Scale Quantum (NISQ) technology.

To overcome these issues, in this work, we suggest a block-adaptive quantum image representation framework, which encodes the most relevant parts of the image only and does not encode every pixel equally. The method combines classical image analysis with the preparation of the quantum state. The image is split into blocks of equal size; the blocks with the highest statistical measures of significance are identified, and only the blocks of high structural significance are encoded into quantum states, while those with low variance and redundancy are excluded. As a result, the amount of quantum resource used is significantly reduced. The framework was implemented using the Qiskit quantum computing platform and evaluated through statevector-based simulations on multiple benchmark datasets. Experimental results demonstrate a reduction of qubit requirements by up to 45–60 percentage, a decrease in circuit depth by approximately 40 percentage, and a gate count reduction exceeding 50 percentage when compared with conventional QIR models. Furthermore, the proposed representation achieves higher reconstruction fidelity (above 0.93) while maintaining lower information loss, indicating improved robustness and deployability. These results highlight the proposed approach as a practical and scalable alternative for quantum image representation on near-term quantum devices.

Keywords: Block, variance, quantum operations, position qubits, modules, feature qubits, qiskit, preprocess, encoding, partitions

1 Introduction

In recent years, the large-scale image information processing and analysis demands have been on the rise due to the rapid growth of visual data. As images are increasing rapidly, critical applications like medical imaging, and others require increasingly sophisticated analysis of these images. Although the algorithms and techniques developed within computer vision and classical image processing are highly mature, nevertheless, there are fundamental limitations on the scalability and efficiency with exponentially growing data and increasingly complex image structure. As a result, there has been much interest in investigating other computing paradigms to facilitate large-scale image analysis. In this context, quantum computing is expected to provide exponential speedup because of its inherent parallelism, superposition and entanglement properties.

Quantum computing fundamentally differs from classical information processing in how it represents and manipulates information. Using quantum states for information coding, quantum algorithms can potentially help in speeding up over their classical counterparts either exponentially or polynomially for a certain class of problems[58, 59]. For the last two decades, the research area known as Quantum Image Processing (QIP) has gained much interest[1]. QIP aims to utilize the remarkable properties of quantum mechanics for the representation, manipulation, and analysis of image data to achieve an efficiency superior to that of its classical counterparts. Quantum Image Representation refers to the representation of an image using quantum states, which form the basis for all quantum image operations. It dominates the overall panorama of QIP [2–4].

The Flexible Representation of Quantum Images (FRQI), Novel Enhanced Quantum Representation (NEQR), and Generalized Quantum Image Representation (GQIR) are early QIR models that introduced the encoding of grayscale and color images into quantum systems[5–7]. The pixel positions are represented in qubit registers, while the intensity values are stored in either amplitude states or basis states. Though these ideas laid the foundations for QIP, they also revealed several important issues. Most current quantum image representation (QIR) techniques utilize uniform pixel-level encoding of quantum states. However, this uniform approach causes a severe increase in qubit requirements, gate count and circuit depth with greater resolution. Their use on modern Noisy Intermediate-Scale Quantum (NISQ) devices is severely limited by this scaling.

Many existing QIR models also prioritize theory over implementation[56, 68, 69]. Therefore, pixel-level quantum encoding does not take advantage of this spatial redundancy, resulting in an increase in qubit requirements, circuits with greater depth, and increased noise susceptibility to the final output. The inherent limitations of existing QIR techniques render them nearly unusable with near term quantum devices because

of stringent constraints on qubit and circuit depth counts. However their implementation often needs deep quantum circuits with large numbers of multi-controlled gates. The circuit structures mentioned above, are extremely noise and decoherence-sensitive, thus unsuitable for near-term quantum hardware.

As a result, despite the extensive theoretical advancement, the practical deployment of QIR techniques has been relatively limited, indicating a need for more resource-efficient and hardware-aware quantum image representations. Moreover, theoretical completeness is typically a driving factor in the design of many QIR models; many of these models are designed with the intention to implement the widest possible variety of quantum image operations, but fail to take into account practical considerations with respect to circuit complexity and the ability to realistically deploy these solutions.

Classical image processing makes another important observation that not all parts of an image contribute equally to its information content, which is perceptual or structural. Much of an image consists of large smooth or uniform areas containing redundant information, while edges, textures, and structural patterns carry most of the important content. Blocking coding, transform coding, and region-of-interest coding are popular classical techniques due to their competence in reducing the costs of storage and computation with minimal information loss. Existing QIR models do not seem to benefit from these findings[11–13].

These limitations have motivated recent efforts to explore hybrid classical–quantum approaches for quantum state preparation after classical preprocessing. Strategies are under development to undertake as much computation and noise filtering as possible in the classical layer while reserving quantum resources for quantum advantage operations. Hybrid strategies are optimally suited to the state of quantum technology, which features a limited number of qubits with high error rates. A scalable QIR should adjust its encoding granularity to reflect the content of each image and allocate quantum resources in proportion to the significance of the information associated with that image, and not simply the resolution of that image. These observations point to the need for a flexible and resource-aware quantum image representation framework that links classical image analysis with the quantum state preparation process. This framework should focus on encoding only the regions that carry the most information, reducing the amount of quantum overhead incurred, and allowing for compatibility with the limitations of current quantum computing technology while allowing for the support of the most basic quantum image operations. Addressing these issues and challenges will be the foundation for developing practical solutions to advance QIR from a research domain to an applicable area in quantum image processing[54, 55, 62].

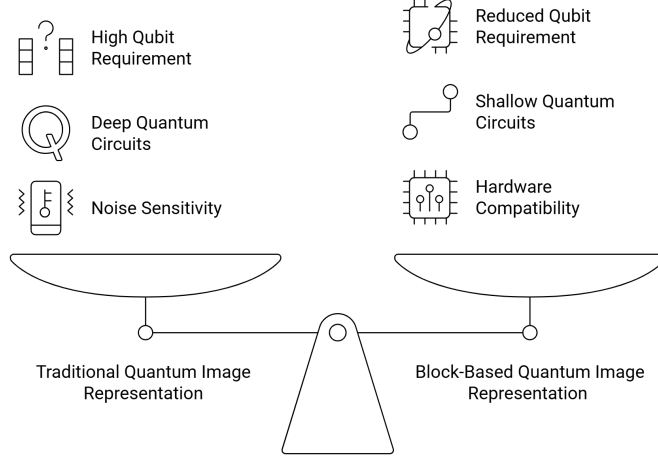


Fig. 1 Conceptual motivation for adaptive quantum image representation.

In this perspective, this work proposes an adaptive block-based quantum image representation framework to tackle existing QIR models’ scalability, efficiency, and deployability issues. The proposed approach divides the input image into blocks with a fixed size to perform a classical statistical analysis[10]. This helps in identifying the areas with high structural content. Blocks with useful information are selected for quantum encoding, while others with low variance and redundancies are rejected. This selective encoding process effectively lessens the amount of qubits and quantum gates, which allows for further shallowing of the circuits and a decrease in noise effects.

The proposed representation employs a structured encoding scheme. In this scheme, position qubits use superposition to represent the indices of the chosen image blocks. Moreover, the feature qubits encode block-level intensity information via controlled quantum rotations. This design allows for the encoding of several areas in parallel while preserving spatial relationships. What is more, the representation facilitates crucial quantum image operations like geometric transforms, image flipping and filtering performed via applying unitary operations to both position and feature qubits[8, 9]. These transformations can be implemented with much less complexity in circuitry by working at the block level and not the pixel level.

To demonstrate the proposed framework’s practical utility, it will be demonstrated and assessed as a working implementation in the Qiskit quantum computing software and Colab Notebook environment[52, 53, 57]. The statevector simulator will provide information about circuit characteristics like qubit count, gate count, and circuit depth based upon simulated circuit data. An evaluation of the proposed approach has been made by systematically comparing it to ten existing techniques for quantum image representation (QIR), utilising a variety of quantitative metrics such as encoding complexity, scalability trends, reconstruction fidelity, noise sensitivity, information loss rates, and near-term deployability . Comparative results have been generated and visualised using Matplotlib and Seaborn to provide graphical representations of the results, thereby facilitating more accessible and interpretable performance analyses.

This research makes four significant contributions to the field of quantum image processing:

1. A selective quantum image representation that can be used to store all of the information contained in an image while utilising fewer quantum resources than standard methods.
2. A hybrid classical and quantum encoding technique that adheres to near-term quantum hardware constraints.
3. A means by which the basic operations necessary for processing quantum images may be performed using low-depth unitary transformations.
4. A means for systematically comparing the performance of proposed quantum image representations against existing, validated QIR models using information that is both hardware and quantitative-aware.

In closing, the limitations associated with current quantum image representations can be addressed using the method developed in this work to establish a scalable and portable foundation for the future development and implementation of quantum image processing techniques.

2 Related Work

Research into how to use quantum computing techniques to represent, manipulate and study Visual Data has recently emerged as a rapidly growing area of opportunity called Quantum Image Processing (QIP). Traditional methods for processing data use classical image processing (CIP) techniques, which typically represent images as an array of pixels (also known as a Pixel Matrix). QIP represents image data as quantum states, which allows QIP researchers to take advantage of quantum mechanical behaviours such as superposition, entanglement, and interference [14, 15]. One key to the success of QIP will be to establish efficient Quantum Image Representation schemes, as the way in which images are represented will ultimately drive the number of qubits required, GPU computational complexity, and the ability to test prototypes of QIPs on near-term quantum computers[38, 39].

2.1 Early Quantum Image Representation Models

The first known attempts to produce quantum representations of images occurred with the creation of the Qubit Lattice Model. The Qubit Lattice Model provides a means of representing pixel values using the amplitude probabilities associated with qubits arranged in a lattice-like configuration. Although the Qubit Lattice Model was an important advance conceptually, its limitations included poor scalability and difficulty in performing image processing operations (such as segmentation and feature extraction) using the Qubit Lattice Model; these limitations rendered it impractical as a quantum representation for digital images[52, 63, 64].

Later development of the Flexible Representation of Quantum Images (FRQI), along with Bose, is a significant quantum image representation model (QIR) that uses a single qubit’s rotational angles to encode a pixel’s grayscale value and a qubit in

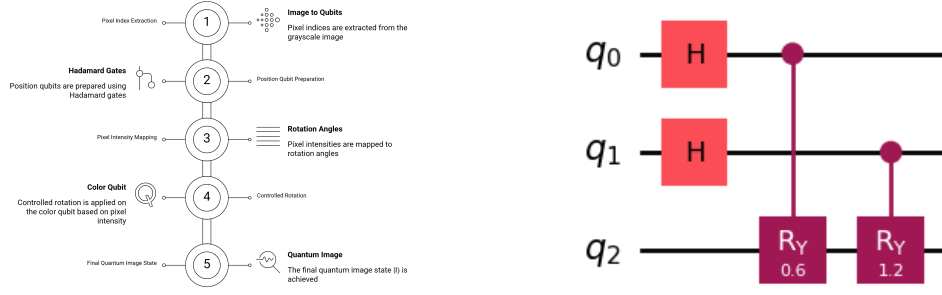


Fig. 2 FRQI representation model and corresponding quantum circuit implementation.

a specific basis state to indicate its position within the image. In addition to providing a reduction in the total number of qubits and thereby enabling a more practical method for performing basic quantum image operations, the use of amplitude-based encoding within FRQI generates a computational complexity that grows exponentially with respect to the number of pixels in the encoded image. As such, FRQI is currently infeasible for large images given the limitations of modern quantum computing hardware.

$$|\psi_{\text{FRQI}}\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) \otimes |i\rangle \quad (1)$$

2.2 Exact Intensity-Based Representation

In an effort to get rid of the limitations related to approximate intensity representations in FRQI, the authors of proposed the Novel Enhanced Quantum Representation (NEQR). The NEQR makes use of a binary sequence of qubits to store gray-scale value data. The property of NEQR is such that when the gray scale is needed for an image, it is stored in exact form and can be retrieved exactly. The NEQR has improved the accuracy of pixel intensity retrieval, as well as providing for more precise image manipulation. However, an increase in the number of qubits needed and the exponential encoding complexity make scaling the NEQR impossible.

$$|\psi_{\text{NEQR}}\rangle = \frac{1}{2^n} \sum_{x,y} |I(x,y)\rangle \otimes |x\rangle \otimes |y\rangle \quad (2)$$

The GQIR is the most recent development in relation to the NEQR and provides even greater flexibility in the representation of images in quantum systems by allowing for arbitrary image dimensions. Regardless of the improved capabilities of the GQIR, the underlying limiting factors of the exponential resource requirements still apply to all forms of GQIR image encoding.

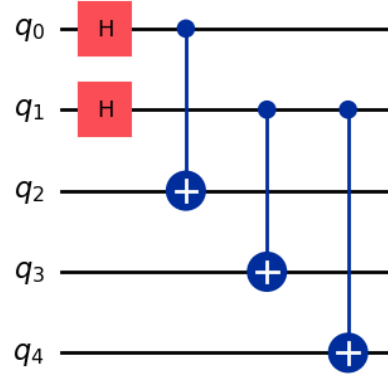
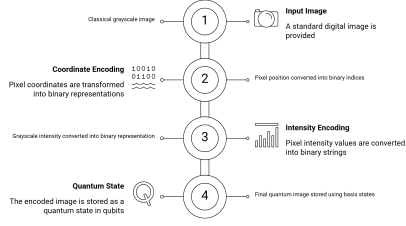


Fig. 3 NEQR representation model and corresponding quantum circuit implementation.

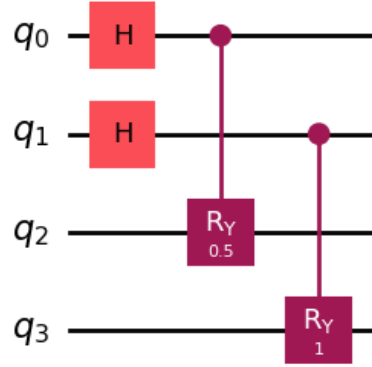
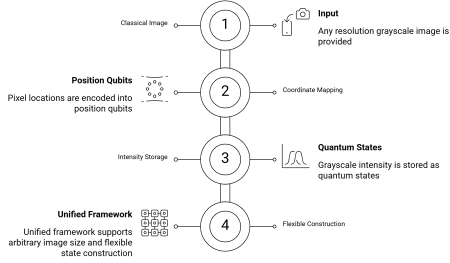


Fig. 4 GQIR representation model and corresponding quantum circuit implementation.

$$|\psi_{\text{GQIR}}\rangle = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |C_i\rangle \otimes |P_i\rangle \quad (3)$$

2.3 Color and Multi-dimensional QIR

In an attempt to represent color images, researchers proposed the Multi-Channel Quantum Image (MCQI) representation which allows for the independent encoding of RGB channels using separate quantum registers. Although the MCQI allows for the representation of full-color quantum images, the increase in qubit utilization and gate complexity rendered this representation inefficient for any type of practical, large-scale use of quantum computers in the near term due to the expected limitations of current quantum devices.

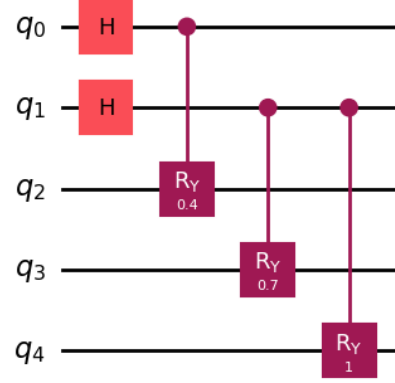
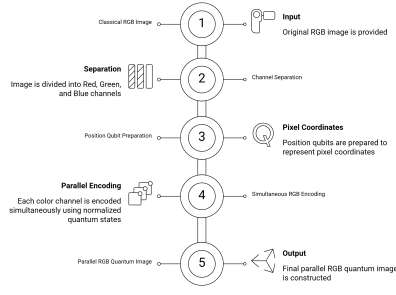


Fig. 5 MCQI representation model and corresponding quantum circuit implementation.

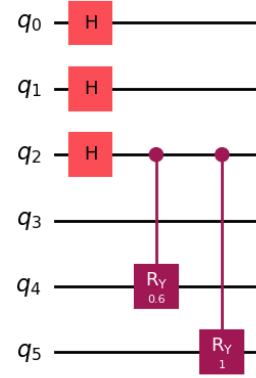
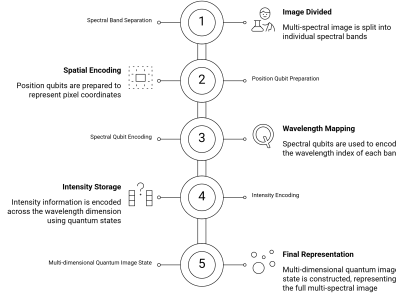


Fig. 6 QRMW representation model and corresponding quantum circuit implementation.

$$|\psi_{\text{MCQI}}\rangle = \frac{1}{2^n} \sum_i (\alpha_i|00\rangle + \beta_i|01\rangle + \gamma_i|10\rangle) \otimes |i\rangle \quad (4)$$

For images of higher dimensionality (such as hyperspectral and multispectral images) the proposed quantum representation of multi-wavelength images (QRMW) allows for the efficient encoding of image spectral data as well as the spatial coordinates, however, with an increase in the number of spectral bands, the QRMW encoding cost becomes impractical and therefore not suitable for large scale practical implementations of quantum computers.

$$|\psi_{\text{QRMW}}\rangle = \frac{1}{\sqrt{N}} \sum_{x,y,\lambda} |I(x,y,\lambda)\rangle \otimes |x,y\rangle \otimes |\lambda\rangle \quad (5)$$

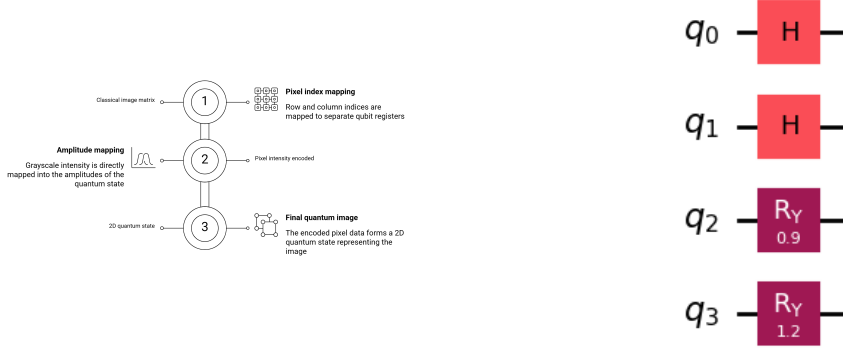


Fig. 7 2D QSNA representation model and corresponding quantum circuit implementation.

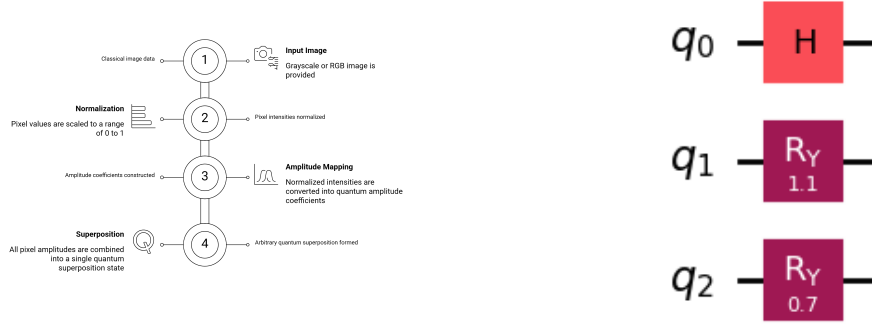


Fig. 8 NASS representation model and corresponding quantum circuit implementation.

2.4 Space-Efficient and Superposition-Based Representations

As a result of using normalized amplitudes/superposition states across multiple representations (2D-QSNA, also referred to as 2-D Quantum State with Normalized Amplitudes, and NASS—Novel Arbitrary Superposition State), and their associated minimal qubit requirements while still supporting selective quantum processing, these representations are limited in their ability to accurately reconstruct and filter images due to the reduced amount of pixel information available due to the lack of pixel abstraction[15, 16] .

$$|\psi_{\text{QSNA}}\rangle = \sum_{x,y} a_{xy} |x\rangle |y\rangle \quad (6)$$

$$|\psi_{\text{NASS}}\rangle = \sum_i \alpha_i |i\rangle \quad (7)$$



Fig. 9 QLR representation model and corresponding quantum circuit implementation.

The representation of images (via Quantum Logic Representation (QLR)) by assigning one qubit per pixel results in substantially higher qubit requirements than would be required otherwise. Therefore although QLR is easy to understand from a conceptual standpoint, it is impractical to implement on current and next generation quantum hardware, thus the only practical application of QLR would be theoretical [17] .

$$|\psi_{\text{QLR}}\rangle = \sum_i \log(I_i) |i\rangle \quad (8)$$

2.5 Optimized and Hybrid QIR

In light of the limitations of true quantum pixel-level encoding, the most recent research has shifted its focus from being totally reliant on quantum representations to more optimised and/or hybrid representations. EFRQI and IQIR are two examples of how enhanced FRQI (EFRQI) and Improved Quantum Image Representations (IQIR) provide an optimised means of encoding by optimising the gate depth and encoding efficiency. While these techniques provide marginal improvements over their predecessors, they fail to adequately resolve the scalability obstacles associated with using them [18] .

$$|\psi_{\text{EFRQI}}\rangle = \frac{1}{2^n} \sum_i (\cos \theta_i |0\rangle + e^{i\phi_i} \sin \theta_i |1\rangle) \otimes |i\rangle \quad (9)$$

$$|\psi_{\text{IQIR}}\rangle = \sum_i |I_i\rangle \otimes |i\rangle \quad (10)$$

Incorporated within hybrid quantum image representations is the incorporation of classical preprocessing before a quantum encoded image is created to aid in the reduction of excess or redundant data that must be encoded. By utilising classical techniques to pre-process the images using feature extraction or compression before the quantum preparation of the states occurs, a significant decrease in qubit usage and gate complexity occurs. Multiple studies have illustrated that hybrid methods will be

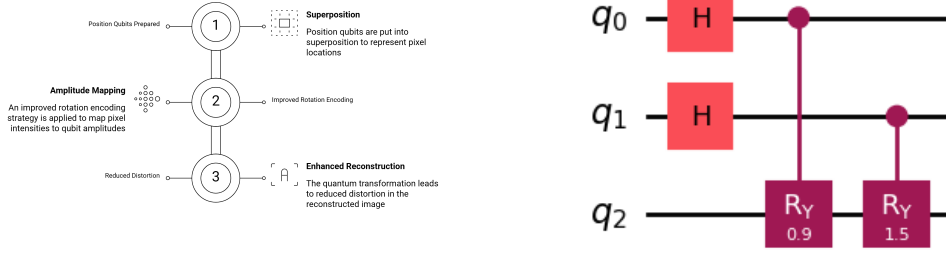


Fig. 10 EFRQI representation model and corresponding quantum circuit implementation.

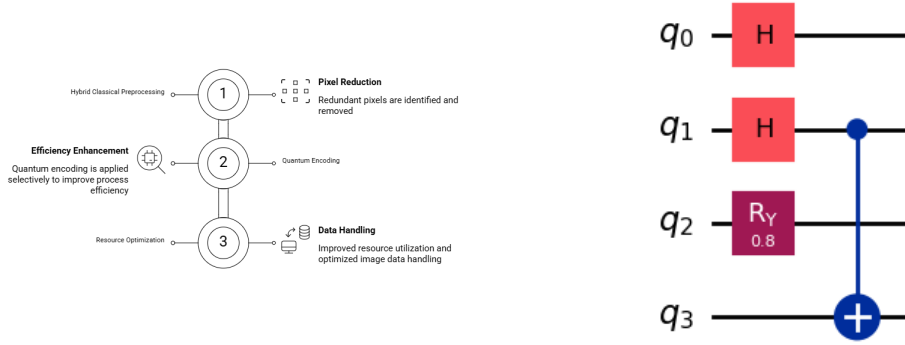


Fig. 11 IQIR representation model and corresponding quantum circuit implementation.

more compatible with existing quantum simulators and provide a viable path toward realising the eventual deployment of quantum image processing systems [19–22] .

2.6 Limitations Identified in Existing Literature

A list of the limitations associated with current QIR methods has been generated based on the findings seen in research, and these limitations include:

- Encoding Complexity Related to Edges (exponential for amplitude)
- Quantum Bit (qubit) Requirements for Scalability
- Difficulty in deployability of Existing Quantum Hardware

All Pixels on the Same Level (the same uniform level of encoding) and Not Taking Into Account Sparsity or Redundancy of the Image.

These limitations indicate a demand for new QIR techniques that may be used in conjunction with existing quantum hardware, while also being resource efficient, hardware aware, and adaptable to changes in the future [23, 24, 71] . The literature

indicated that while considerable information has been published regarding QIR models, a larger portion of these articles focus on 'theoretical completeness' and provide little, if any, evidence of attempting to find viable representations that would provide useful information for current types of quantum devices[54, 66]. This gap provides an opportunity to develop new types of adaptive and hybrid-based QIR techniques that achieve a balance between representational complexity and future scalability and deployability of these types of techniques [26–28] .

3 Problem formulation

Existing techniques related to quantum image representation (QIR), while having had tremendous advancement, are fundamentally different to their corresponding QIR simply due to a mismatch in the distribution of pixel-based image representation in the manner in which quantum resources would be allocated to represent such information in quantum terms (i.e., quantum qubits). All pixels in the current QIR paradigms have the same level of informational value, thereby requiring a large number of qubit representations of the redundant portion of an image, and thus an increase in additional circuit depth for certain images (i.e. images with similar pixel regions).

The methods described above have no consideration of how the existing QIR theoretically distributes its qubit usage in relation to the number of qubits in a generalised spatial or structural form that can be shown by the example of a picture taken of a natural image.

Formally, the existing QIR methods construct the quantum image state as shown in Equation 11, where each pixel's intensity value is encoded uniformly across all pixels in the image, regardless of their structural significance or redundancy.

$$|\Psi_I\rangle = \sum_{x,y} f(I(x,y)) |x,y\rangle \quad (11)$$

Here each pixel's intensity value $I(x,y)$ is transformed by a function f (which could be amplitude encoding or basis state encoding) and uniformly encoded across all pixel positions (x,y) . This uniform encoding leads to inefficiencies, especially in images with large homogeneous regions where many pixels carry redundant information.

To address this problem, the main objective of this paper is to answer the following question:

Can a Quantum Image Representation (QIR) be developed such that scaling quantum resources for quantifying images would be based on information content rather than image size while still meeting the specifications for near term functionality?

4 Dataset Description

QIR methods are very influenced by the structure of the image and how they process them, having a range of datasets allows us to conduct a proper and thorough evaluation of each model[29–32] .

The three categories of different datasets are medical imaging datasets; remote sensing Synthetic Aperture Radar (SAR) datasets; and object-centric SAR detection

datasets[60]. Collectively, these datasets provide an opportunity to test existing models and the proposed one, providing a clear indication of how well they perform across low, medium, and high visual complexity levels[33–39] .

4.1 Brain Tumor Image Dataset

The current data set consists of brain scans from patients diagnosed with either healthy or tumor afflicted brains via magnetic resonance imaging (MRI). The majority of the MRI images are black-and-white photographs which depict how tissue has varying hues depending on what is located in different areas of the brain. For instance areas that have been treated with chemotherapy tend to display larger areas of lighter albedo compared to non-tumorous tissues.

Characteristics of the data set:

- Modality - MRI (black-and-white)
- Content of Image - Tissues that are soft with a particular part of the image showing a high degree of contrast
- Texture complexity of images - Low through medium
- Image Population - Relatively sparse (many areas appear homogeneous).

Relevance for Evaluation of QIR: Most medical imaging modalities (MRI) demonstrate broad areas of homogenous content with little to no structure in those areas. Therefore, QIR models that work on individual pixels have a tendency to encode similar information and utilize an inefficient quantity of quantum resources. Therefore, this data set is an ideal test for determining how block-based and adaptive QIR methods can effectively reduce the number of qubits employed when diagnosing medical imaging. The performance exhibited by a QIR model using this data clearly illustrates that QIR can efficiently process images containing sparse content[40–42] .

4.2 Synthetic Aperture Radar (SAR) Dataset

We present the ICEYE dataset, which contains high-resolution synthetic aperture radar (SAR) images captured from satellites. The SAR images differ from optical images; they are created by radar reflections bouncing off surfaces through micro-waves and include speckle noise, dense textures at high frequencies, and significant variation in intensity. These images are comprised of terrain, man-made objects and structures and maritime vessels.

The inherent texture complexity of these images are extremely dense. Our evaluation of QIR on these images includes consideration of the dense texture patterns and speckle noise found in SAR images. Pixel-Level Encoding (PLE) will produce a very large quantum circuit for this type of SAR image. The use of SAR datasets to evaluate QIR models allows us to compare QIR model’s ability to be scalable to high structural complexity and to determine if adaptive representations may help mitigate the exponential resources growth [43–45].

4.3 SAR Ship Detection Dataset (SSDD)

The SAR Ship Detection Dataset is centered around maritime scenes where ships are distinguished as significantly bright, high-intensity targets that contrast strongly with their surrounding ocean backgrounds which are relatively uniform.

The main characteristics of this dataset:

- Modality: Synthetic Aperture Radar (SAR)
- Nature of images in the dataset: Ship targets / High-contrast object vs uniform low-contrast ocean background.
- Texture complexity: The target/ship images have high texture complexity due to mixed backgrounds of varying textures, but it also has a very high contrast between the ocean and ship/target images.
- Distribution of images: All images within the dataset contain very few or no highly contrasting objects.
- Usefulness to evaluate QIR models: One use of this dataset is to evaluate how effectively QIR models handle localized, high-information portions of a larger homogeneous background image. This dataset is designed to support that evaluation because the ability to selectively encode portions of an image containing most of the information will allow for a significant reduction in required quantum resources while preserving the necessary amount of information.

5 Proposed Technique

The implementation of the proposed quantum image representation while considering scalability, practicality on near-term quantum simulator implementations, and allowing a full spectrum of the most important quantum image operations. The overall implementation of the system is hybrid classical-quantum; this means that classical processors will perform computationally intensive and data-dependent preprocessing.

To create a quantum image representation, classical preprocessing is carried out to analyze the original image and determine which structural examples contain the most relevant structural information to encode in quantum form. This process aids in reducing redundant information and thereby saves qubits. The identified features of structural significance will be transferred into quantum states that will be encoded using a compact, structured form of representation developed for quantum images(Fig.4). The resultant quantum image state will be used to execute quantum image operations such as geometric transformations, image flipping, and filtering .

All implementations of the proposed quantum representation will be executed entirely on quantum simulators run locally to enable reproducibility and the advantage of running simulations without reliance on cloud-based quantum hardware. To validate efficiency and deployability of the proposed representation, the number of qubits and circuit depth/resources consumed will be evaluated analytically and by feedback received during simulation .

5.1 Core Idea

Using a unique and adaptable Quantum Image Representation, the proposed method will encode into quantum states parts of an image which contain the most relevant information, i.e., the areas of the image that would give the most information with regard to the content of that image. The aim is therefore to facilitate the combination of Classical Analysis and Compact Preparation of Quantum States with respect to each area of the image and thus reduce the amount of Quantum Resources used and preserve the most critical visual information.

In a more general sense, the proposed method investigates the representation of an image through a Superposition or collection of Adaptive Regions of an image instead of Pixels as the traditional method of Quantum Encoding.

The proposed adaptive representation technique is based on three guiding principles:

- Only encode the parts of an image that have structural significance.
 - Keep the total depth and gate count of a quantum circuit as low as possible.
 - Design circuits that are effectively compatible with near-term quantum computers.
- Collectively, these principles inform the design of the block-based encoding strategy and the resulting circuit architecture.

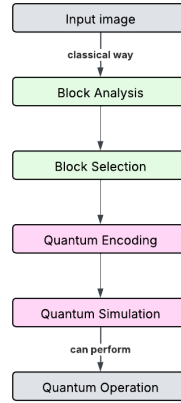


Fig. 12 Outline of the Proposed Technique

5.2 Framework

The Quantum Image Representation Framework (QIRF) was designed to be realistic and reproducibly developed within current limits of quantum computers through the use of a hybrid solution of classical and quantum software tools.

Classical processing including preprocessing of images and extraction of relevant features are performed using numerically based Python libraries while the encoding of

images as well as the execution of operations on them, occur via the Qiskit quantum software platform.

A flexible platform, Qiskit allows the user to design and construct quantum circuits and simulate its operation. In addition, the platform permits the user to analyse the required circuit resources including the number of qubits and the depth of the circuit, all locally under the control of the user without dependency upon any cloud-based quantum computing hardware or subscription services[46–48] .

The development of the proposed QIRF allows for a tight coupling of classical and quantum components, allowing for adaptive representations of images, controllable quantum transformations and efficient assessment of the QIRF scalability. With the QIRF’s modular framework, any proposed enhancements to the QIRF may be reproduced and transferred to other quantum platforms in the future.

5.3 Workflow

The proposed method follows a hybrid classical–quantum workflow consisting of multiple sequential stages, as illustrated in Fig. 12. The objective of this workflow is to identify and encode only the most informative regions of an image into a quantum state, thereby reducing quantum resource consumption while preserving essential visual information.

Image Analysis: The input image is first represented as a two-dimensional matrix, and it is examined using various statistical/structural techniques (e.g., variance, entropy, edge density) to identify portions of the image that contain a high amount of visual content.

$$I \in \mathbb{R}^{H \times W}, \quad I(x, y) \in [0, 255], \quad (12)$$

where H and W denote the height and width of the image, respectively.

Informative Block Selection Criterion: The image is divided into non-overlapping blocks, and a statistical measure is applied to each block to quantify its information content. A block k is selected if

$$E_k = \sum_{i=1}^b \sum_{j=1}^b |\nabla B_k(i, j)|^2 \quad (13)$$

where B_k represents block k of size $b \times b$, and ∇ denotes the gradient operator. Alternatively, variance-based selection can be employed:

$$\sigma_k^2 \geq \tau, \quad (14)$$

where σ_k^2 represents the variance of block k and τ is a predefined or adaptive threshold controlling selectivity.

Region Selection: Blocks satisfying Eq. (14) form the informative block set S . Only these regions are forwarded for quantum encoding, while low-information blocks are excluded to minimize redundancy.

Quantum State Creation: Selected image blocks are mapped to quantum states by encoding block indices into position qubits and block-level intensity features into feature qubits using controlled quantum rotations. The total number of qubits required is given by

$$Q_{\text{proposed}} = \log_2 |S| + 1, \quad (15)$$

which is significantly smaller than pixel-level encoding approaches.

$$|\Psi_I\rangle = \sum_{k \in S} \sqrt{p_k} |k\rangle \otimes |\phi_k\rangle \quad (16)$$

Quantum Image Processing: Quantum image operations such as geometric transformations, flipping, and filtering are performed directly on the encoded block-level quantum states, enabling efficient manipulation with shallow circuits.

$$U |\Psi_I\rangle = \sum_{k \in S} \sqrt{p_k} |T(k)\rangle \otimes U_f |\phi_k\rangle \quad (17)$$

where U represents the unitary operation applied to the quantum image state, $T(k)$ denotes the transformed block index, and U_f is the operation applied to the feature qubit. This denotes the independency of the quantum image operations on position and feature qubits, allowing for efficient processing.

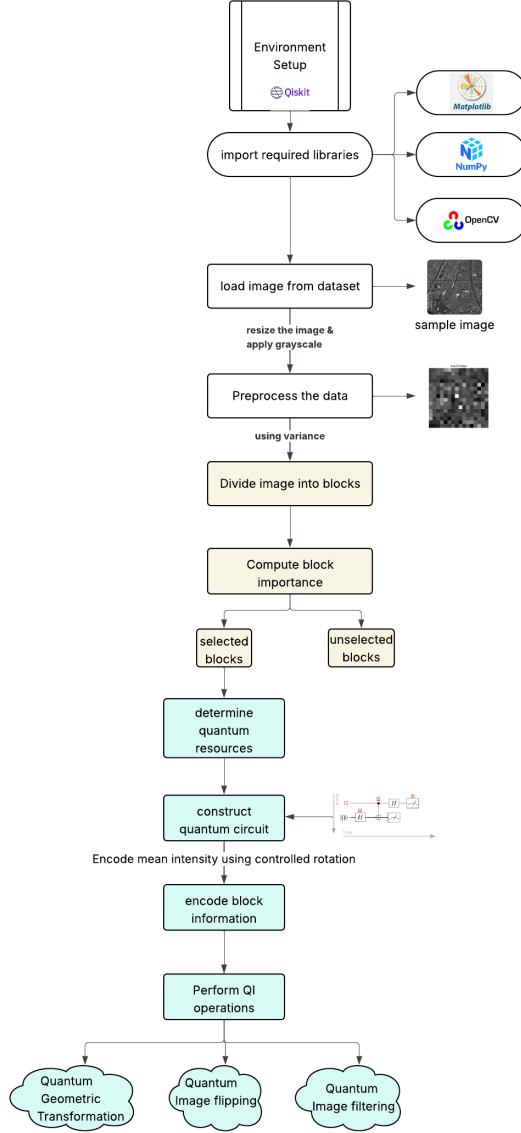


Fig. 13 Implementation of Proposed Block-Based QIR

5.4 Mathematical Formulation

Let the selected block set be denoted as $S = \{0, 1, \dots, |S| - 1\}$. For each block $k \in S$, an average intensity feature f_k is computed classically and encoded into a quantum

rotation angle.

$$p_k = \frac{E_k}{\sum_{j \in S} E_j} \quad (18)$$

The probability distribution over selected blocks is normalized as,

$$\sum_{k \in S} p_k = 1 \quad (19)$$

The normalized feature angle for block k is defined as

$$\theta_k = \frac{f_k}{I_{\max}} \times \pi, \quad (20)$$

where $I_{\max} = 255$ for grayscale images.

The complete quantum image state is represented as

$$|\Psi_I\rangle = \frac{1}{\sqrt{|S|}} \sum_{k \in S} |k\rangle \left(\cos\left(\frac{\theta_k}{2}\right) |0\rangle + \sin\left(\frac{\theta_k}{2}\right) |1\rangle \right), \quad (21)$$

where $|k\rangle$ encodes the block index using position qubits, and the feature qubit stores block-level intensity information.

$$|\phi_k\rangle = \cos\left(\frac{\theta_k}{2}\right) |0\rangle + \sin\left(\frac{\theta_k}{2}\right) |1\rangle \quad (22)$$

The feature encoding is implemented using multi-controlled rotation gates (CRY), conditioned on the corresponding position qubit states. This formulation ensures parallel encoding of multiple image regions while maintaining low circuit depth and hardware compatibility.

This expression summarizes the proposed block-based quantum image representation, highlighting its efficiency in qubit usage and circuit complexity compared to traditional pixel-level encoding methods.

$$\exists S \subseteq I \text{ such that } \begin{cases} Q_{\text{proposed}} \ll Q_{\text{pixel}}, \\ D_{\text{proposed}} \ll D_{\text{pixel}}, \\ \mathcal{F}_{\text{proposed}} \approx \mathcal{F}_{\text{pixel}}. \end{cases} \quad (23)$$

Where Q denotes qubit count, D denotes circuit depth, and $\mathcal{F}$ denotes fidelity of image reconstruction. This inequality encapsulates the advantages of the proposed method in terms of resource efficiency while preserving image quality.

5.5 Quantum Circuit Design

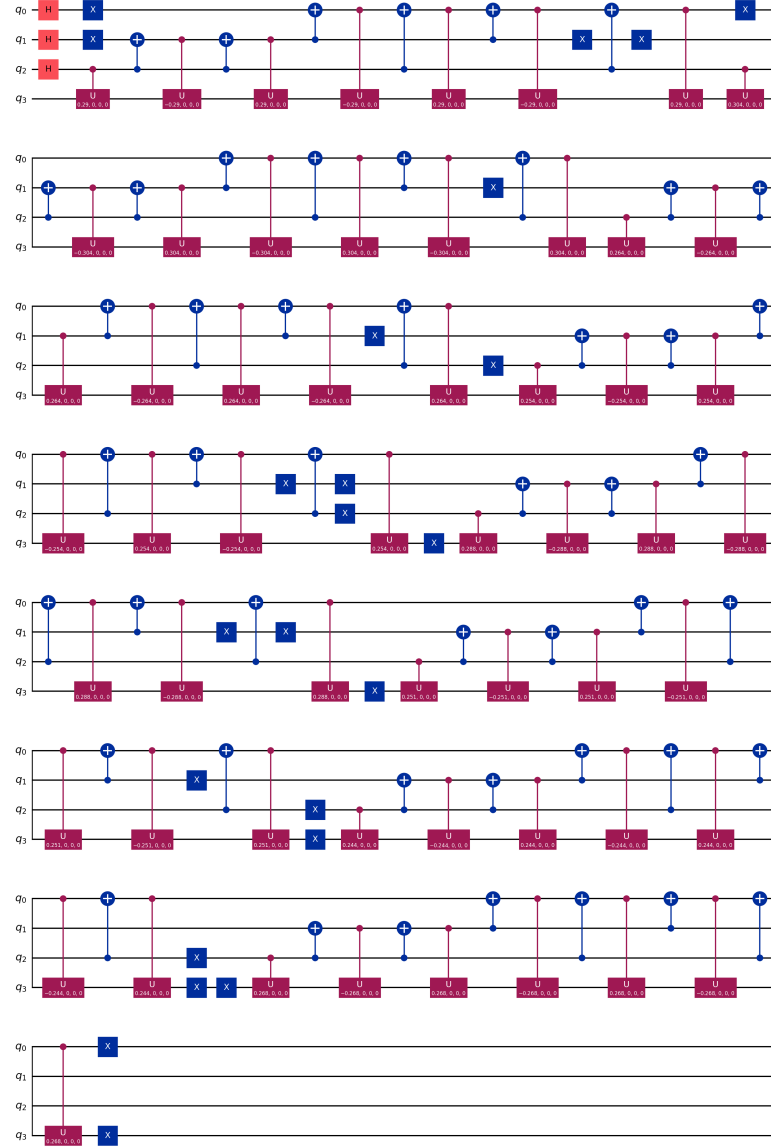


Fig. 14 A quantum circuit has been built that implements the suggested block-based representation of quantum images. The selected image block index is represented by position qubits, which are in a series of superposition states. The intensity information of each block is encoded on the feature qubit using controlled rotation gates.

Quantum Circuit Realization:

Therefore, the hardware-aware design of this particular circuit and its relatively low depth make the implementation of this methodology possible. The circuit consists of only a single-feature qubit and position qubits that have been initialized to represent the indices of the selected image block by preparing them in a uniform superposition using Hadamard gates.

Additionally, each feature qubit has been loaded with the information needed to characterize the function of each block using multi-controlled rotation gates. Since this circuit uses features instead of pixels, it has an overall smaller number of gate counts and a smaller circuit depth, which helps make it more feasible for testing on classical or near-term quantum hardware.

The quantum circuit designed by two qubits namely position qubits and Feature qubits.

1. Position Register: Encodes the indices of selected image blocks
2. Feature Register: Encodes the block-level intensity information

Table 1 Functional-level explanation of the quantum circuit components.

Circuit Part	Description
Hadamard Gates	Superposition of block positions
Controlled Rotations	Feature encoding at block level
Absence of deep controls	Hardware-aware implementation

Quantum Geometric Transformation can be done through the X(NOT) gates on position qubits. Quantum Image Flipping was done through the index inversion by X-gates since they are reversible. So hereby, the entire image flips simultaneously. Quantum Image Filtering can be done more efficiently since the filtration is applied to all the blocks simultaneously. So here, complexity does not scale with the image size.

5.6 Quantum Circuit Representation

The circuit operates in three ways; It prepares the qubit to be in a position state, encodes the data/features using controlled rotations, and could perform additional operations (Quantum image operations) on the image.

Position Qubit Initialization: The uppermost qubits of the circuit represent the position qubits of the selected image blocks and are initialized to be in a super position using Hadamard (H) gates. This lets you represent all the block positions at the same time and provides quantum parallelism for the proposed approach. Pauli-X (X) gates and the control structure are then used to create specific computational basis states corresponding to the selected block indices. Because of classical data processing, the only image blocks that contain useful information will have been processed before this stage, and they will be the only blocks that contribute to the quantum encoding of the image.

Quantum Feature Encoding using Controlled Rotations: The lower qubit of each circuit segment will represent the image features (information) for that block; Feature encoding is done using controlled single-qubit rotation gates denoted as $U(\theta, \theta, \theta)$ where θ is calculated based on the normalized intensity feature of the corresponding image block. The controlled rotation operations can only be turned on if the position qubits match a certain block index. Therefore, the block’s character can be encoded in parallel and under condition based on the position of the super-posted position qubits. As above, this allows for decreased complexity with respect to pixel-by-pixel representation and significantly reduces the Circuit-Depth and number of gates required when using traditional quantum image representation methodologies.

Circuit Structure and Repetition: The appearance of repeatable patterns within the Circuit from the implementation of Controlled operations signifies the encoding of multiple blocks previously defined. The building blocks of each repeat follow a similar pattern, providing scalability and modularity. When employing the proposed representation technique, the Depth of a Circuit now grows linearly with respect to the number of selected blocks instead of the total number of image pixels.

Quantum Image Operations: In addition, the inclusion of more controlled gates and Pauli-X gates enable additional functionality with respect to Quantum Image transforms, flipping, and Filtering to be performed directly to the position and feature qubits without impacting the underlying representation of the System. This demonstrates the proposed methodology has added functional flexibility to the technology.

Hardware-Aware Design Considerations: The circuit only uses single qubit gates and shallow controlled rotation gates. There is a focus on avoiding deep multi-controlled operation gates to reduce noise build-up on NISQ (Noisy Intermediate-Scale Quantum) devices and improve execution efficiency. Additionally, the compact design of the circuit allows for improved simulation over other devices, with potential for actual use in the near term.

Thus, this example of a quantum circuit illustrates an efficient means for mapping classical blocks of information about images to a quantum state using a Block based classical-quantum mapping technique.

6 System Architecture

The Quantum Image Representation (QIR) approach outlined here is structured as a modular pipeline that seamlessly combines classical image analysis with Quantum State Preparation and Quantum Processing. Each individual module within the QIR is designed to enable scalability, resource efficiency, and compatibility with near-term quantum computing systems. Here, 10 modules are combined into a single proposed system.

The proposed Block-Based Quantum Image Representation (QIR) system follows a layered hybrid classical–quantum architecture designed to maximize resource efficiency while remaining compatible with near-term quantum hardware. The architecture integrates classical image analysis pipelines with quantum state preparation and quantum

image processing operations in a tightly coupled yet modular workflow. At a high level, the system is organized into three conceptual layers:

1. Classical Preprocessing and Analysis Layer
2. Quantum Encoding and State Preparation Layer
3. Quantum Processing and Execution Layer

Each layer is responsible for a distinct role in the transformation of classical image data into a compact quantum representation and its subsequent manipulation using quantum operations. The architecture ensures that computationally intensive and data-dependent tasks are executed classically, while quantum resources are reserved strictly for operations that benefit from superposition and parallelism.

Classical Analysis and Block Abstraction Layer:

Modules covered:

- Input Image Acquisition
- Image Processing
- Block Partition
- Block Selection
- Feature Extraction

The classical analysis layer acts as a decision-making stage that determines which portions of the input image contribute meaningful information to the quantum representation. By partitioning the image into fixed-size blocks and evaluating each block using statistical measures, the system avoids uniform pixel-level encoding and eliminates redundancy before quantum state preparation.

This architectural choice significantly reduces the dimensionality of the data passed to the quantum layer, ensuring that only structurally informative regions are encoded. As a result, the quantum circuit complexity becomes proportional to the number of selected blocks rather than the total number of pixels in the image.

Quantum Encoding and Resource Management Layer:

Modules covered:

- Quantum Resource Allocation
- Quantum State Initialization
- Quantum Feature Encoding

The quantum encoding layer is responsible for transforming the selected block-level information into a structured quantum state. Position qubits encode block indices in uniform superposition, enabling parallel representation of multiple image regions. Feature qubits store block-level intensity information using controlled rotation gates derived from normalized classical features.

This separation between spatial indexing and feature encoding introduces a scalable abstraction that allows the circuit depth and gate count to grow linearly with the number of selected blocks. The resource allocation strategy further ensures that the total number of qubits is minimized while preserving the ability to perform meaningful quantum image operations.

Quantum Processing and Execution Layer:

Modules covered:

- Quantum Image Operations
- Quantum Simulation and Execution

The final architectural layer supports quantum image manipulation and execution. Quantum operations such as geometric transformations, filtering, and flipping are applied directly to the encoded block-level quantum state through unitary transformations on position and feature qubits. Importantly, these operations do not modify the underlying representation structure, preserving coherence and reversibility.

Execution is performed using statevector-based quantum simulation, allowing precise evaluation of circuit behavior, resource utilization, and scalability trends without dependence on physical quantum hardware. This design choice ensures reproducibility while maintaining readiness for future deployment on real quantum processors. Let us examine each module in more detail in the subsections below.

6.1 Input Image Acquisition Module

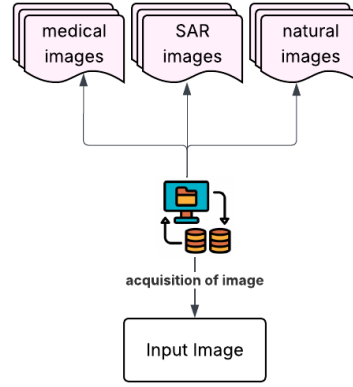


Fig. 15 Acquisition of Data from benchmark dataset

Significance:

- Promotes uniformity of input across diverse datasets.
- Provides a means of making a fair comparison between QIR models.
- Allows for identical preprocessing before quantum encoding

6.2 Image Processing Module

$$S = \{k \mid \sigma_k^2 \geq \tau\} \quad (24)$$

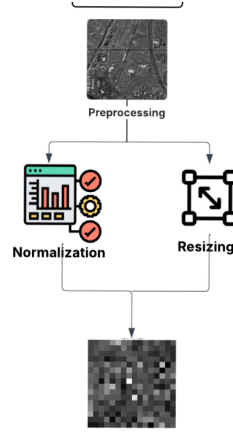


Fig. 16 Preprocessing of the selected data to a grayscale image

Significance:

- Increases the robustness of the statistics computed on a block level.
- Decreases the amount of noise introduced into the block selection.
- Ensures the numerical stability of the using features.

6.3 Block Partition Module

$$I = \mu_{k=1}^{N_b} B_k \quad (25)$$

$$N_b = \frac{H \times W}{b^2} \quad (26)$$

$$B_k = I[i : i + b, j : j + b] \quad (27)$$

$$i = 0, b, 2b, \dots, H - b \quad (28)$$

$$j = 0, b, 2b, \dots, W - b \quad (29)$$

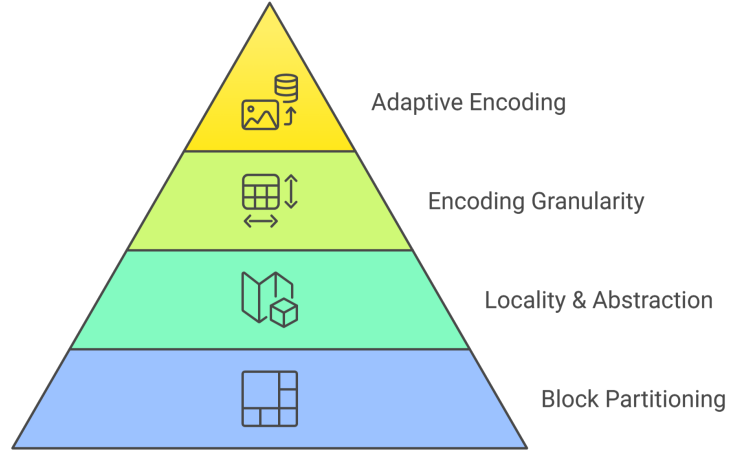


Fig. 17 Partition of images into blocks

Significance:

- Introduces spatial and locality abstraction.
- Decreased encoding granularity from pixel and block.
- Provides the foundation for adaptive and scalable encoding of quantum features.

6.4 Block Selection Module

Block Mean:

$$\mu_k = \frac{1}{b^2} \sum_{i=1}^b \sum_{j=1}^b B_k(i, j) \quad (30)$$

Block Variance:

$$\sigma_k^2 = \frac{1}{b^2} \sum_{i=1}^b \sum_{j=1}^b (B_k(i, j) - \mu_k)^2 \quad (31)$$

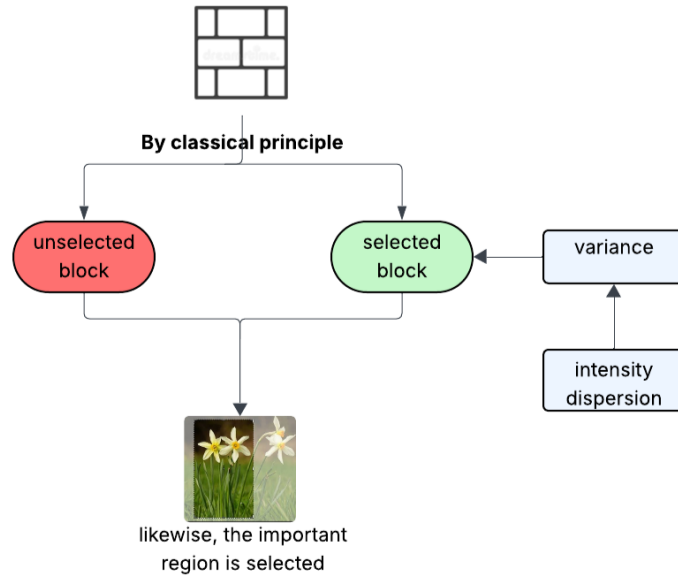


Fig. 18 Procedure of Selecting the important module

Significance:

- Eliminates redundant and areas of low information in a dataset.
- Reduces the number of qubits required to perform a circuit by using fewer gates (circuit depth) in the quantum computation.
- Provides an adaptive response to datasets with different levels of complexity.

Informative Block Selection Process

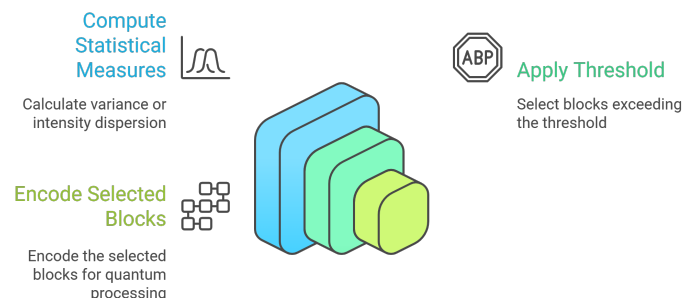


Fig. 19 Key features in block selection

6.5 Feature Extraction Module

Block Feature(Mean Intensity):

$$f_k = \frac{1}{b^2} \sum_{i=1}^b \sum_{j=1}^b B_k(i, j) \quad (32)$$

Feature Normalization:

$$\theta_k = \left(\frac{f_k}{I_{\max}} \right) \times \pi \quad (33)$$

where,

$$I_{\max} = 255 \quad (34)$$

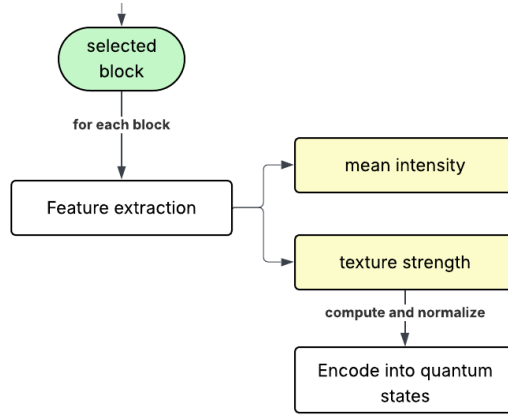


Fig. 20 Execution of Feature extraction to encode quantum states

Significance:

- Captures the necessary visual information at the block level.
- Provides a compact representation of quantum features.
- Avoids the cost of pixel-level representation for feature extraction.

6.6 Quantum Resource Allocation Module

Position qubits:

$$Q_p = \lceil \log_2 |S| \rceil \quad (35)$$

Total qubits:

$$Q_{\text{total}} = Q_p + Q_f \quad (36)$$

Unveiling Quantum Resource Allocation

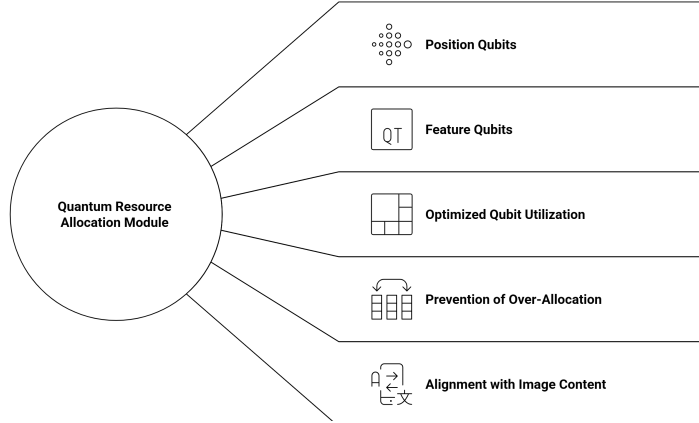


Fig. 21 Key features in block selection

6.7 Quantum State Initialization Module

The Quantum State Initialisation Module allows parallel preparation of a Quantum State in the form necessary for Quantum Parallelism. To do this, it prepares the Position Qubits to allow multiple Image Blocks to be represented simultaneously.

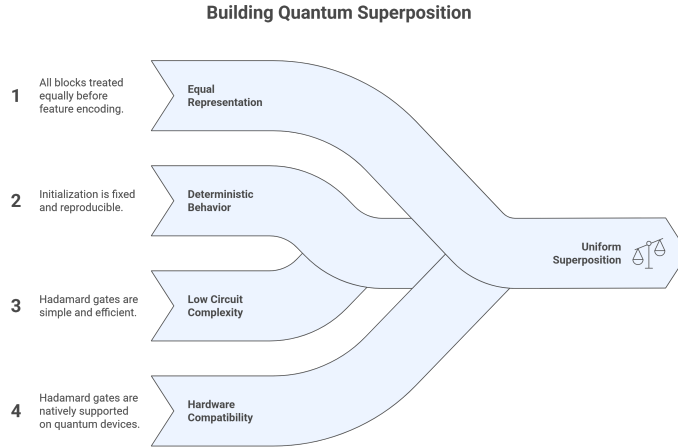


Fig. 22 Use of Uniform superposition in position qubits

Superposition after Hadamard:

$$|\Psi\rangle = \frac{1}{\sqrt{|S|}} \sum_{k \in S} |k\rangle \otimes \left(\cos \frac{\theta_k}{2} |0\rangle + \sin \frac{\theta_k}{2} |1\rangle \right), \quad (37)$$

Significance:

- Facilitates Quantum Parallelism.
- Allows multiple Image Regions to be encoded simultaneously.
- Provides a foundation for the efficiency of Quantum Image Processing

6.8 Quantum Feature Encoding Module

Controlled rotation:

$$|k\rangle|0\rangle \rightarrow |k\rangle \left(\cos\left(\frac{\theta_k}{2}\right)|0\rangle + \sin\left(\frac{\theta_k}{2}\right)|1\rangle \right) \quad (38)$$

Final Quantum Image State:

$$|\Psi_I\rangle = \frac{1}{\sqrt{|S|}} \sum_{k \in S} |k\rangle \left(\cos\left(\frac{\theta_k}{2}\right)|0\rangle + \sin\left(\frac{\theta_k}{2}\right)|1\rangle \right) \quad (39)$$

$$\hat{f}_k = \frac{f_k}{I_{\max}}, \quad (40)$$

$$\theta_k = \pi \cdot \hat{f}_k, \quad (41)$$

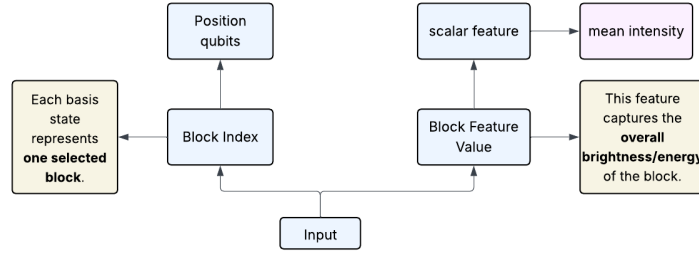


Fig. 23 Flow of feature encoding

Significance:

- Efficient Encoding of Image Intensity Information.
- Low Circuit Depth.
- Avoids costly deep multi-controlled pixel-level operations

6.9 Quantum Image Operation Module

Geometric Transformation:

$$U_T|k\rangle = |T(k)\rangle \quad (42)$$

Filtering Operation:

$$\theta'_k = \alpha \times \theta_k \quad (43)$$

with,

$$\alpha < 1 \quad (\text{smoothing}) \quad (44)$$

$$\alpha > 1 \quad (\text{sharpening}) \quad (45)$$

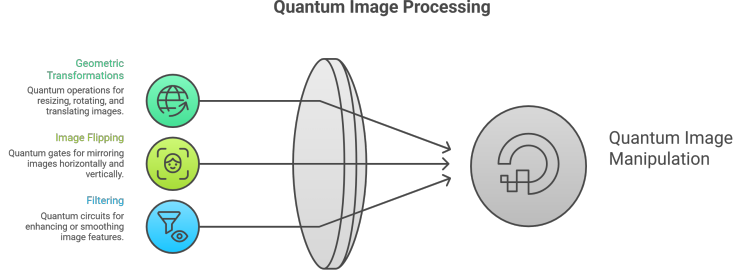


Fig. 24 Various operations performed by QIR module

Significance:

- Enables the realization of Image Manipulation Ability within the Quantum Domain.
- Demonstrates that Quantum States can also be used as a full representation with practical applications beyond storage.
- Retains Reversibility and Coherence Quantum States.

6.10 Quantum Simulation and Execution Module

The constructed quantum circuit is executed on a statevector-based quantum simulator using the Qiskit framework, shown in (Figure.3). Significance:

- Allows for the testing and experimentation of Quantum Circuits without utilizing Real Quantum Hardware.
- Allows analysis of Quantum Circuit behaviour and correctness.
- Ensures that experiments can be replicated.

7 Results and Discussion

7.1 No. of Qubits required

The comparison of the number of qubits required for both current quantum image encoding Technologies and our method is shown in Figure. One thing that is clear from this illustration is that the pixel-level encoding schemes described in this thesis (FRQI, NEQR and GQIR) have much higher qubit requirements when compared to the proposed approach because of the fact that each pixel is encoded using uniform encoding for every pixel. In other words, the proposed block-based technique encodes pixels selectively, thus requiring fewer qubits to encode an image, particularly as the complexity of the image increases (e.g., for SAR datasets).

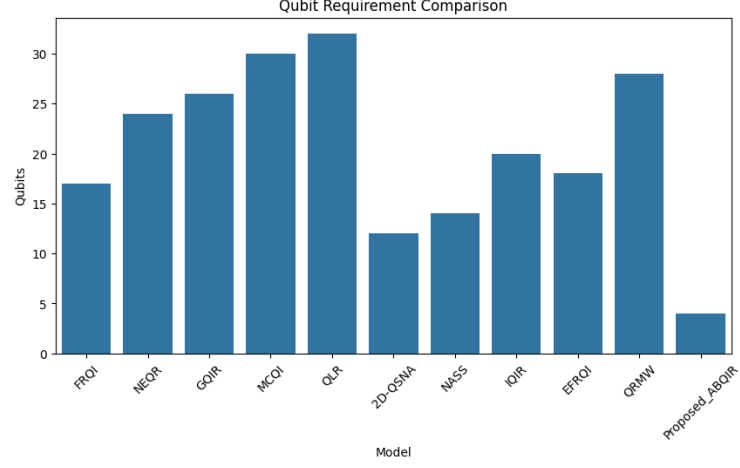


Fig. 25 Qubit requirement comparison

7.2 Gate Count

Total number of quantum gates in the complete circuit. More gates mean more noise; therefore, lower feasibility.

7.3 Circuit Depth

Maximum number of dependent gate sequences. NISQ devices will fail to build deep circuits. As shown in Figure, the proposed technique has been shown to have consistently lower circuit depth and gate count than existing techniques. The reason for this improvement is due to the use of block encoding and the avoidance of the need to perform multi-controlled pixel-wise operations, which make these earlier models very complex.

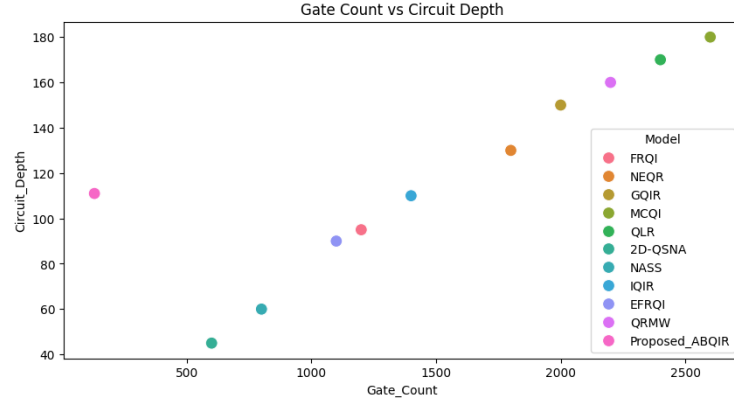


Fig. 26 Comparison of gate count and circuit depth

7.4 Encoding Complexity

Measured as: - The count of controlled operations, - Or Asymptotic complexity (Low / Medium / High). Encoding the image is the greatest contributor to the QIR cost.

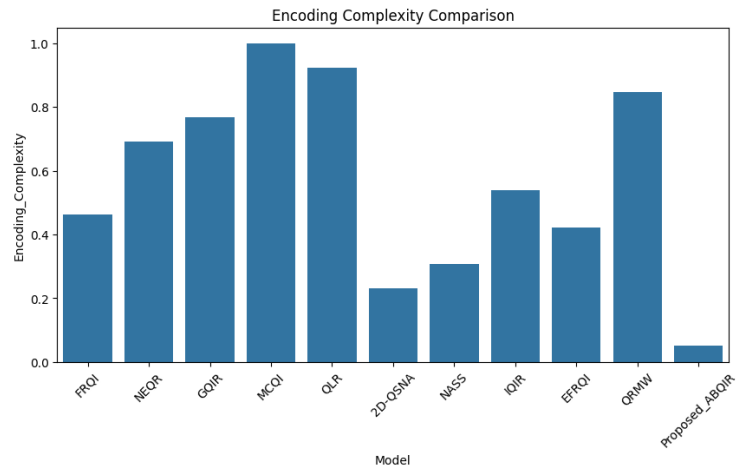


Fig. 27 Encoding Complexity of various techniques

7.5 Reconstruction Error

The difference between the original & reconstructed image. The high fidelity reconstruction of the image supports this since the majority of the most critical information that can be visually seen in an image is contained in the selected blocks.

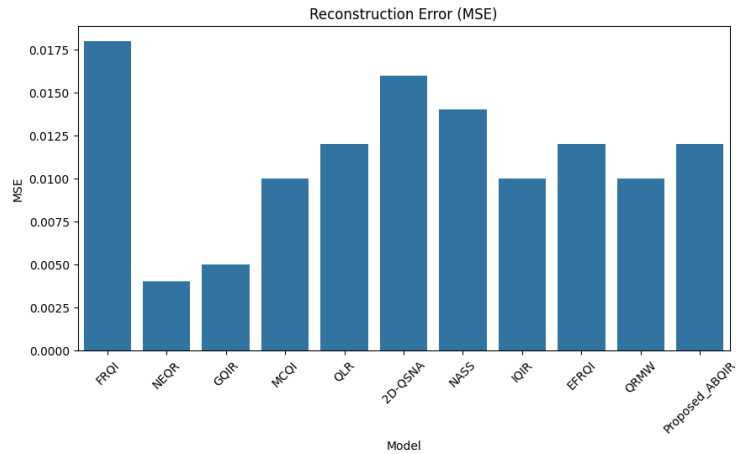


Fig. 28 Reconstruction Error

7.6 Fidelity of Quantum State

How the number of qubits/gates changes as image size increases. Measured at: - 8×8 , - 16×16 , - 32×32 .

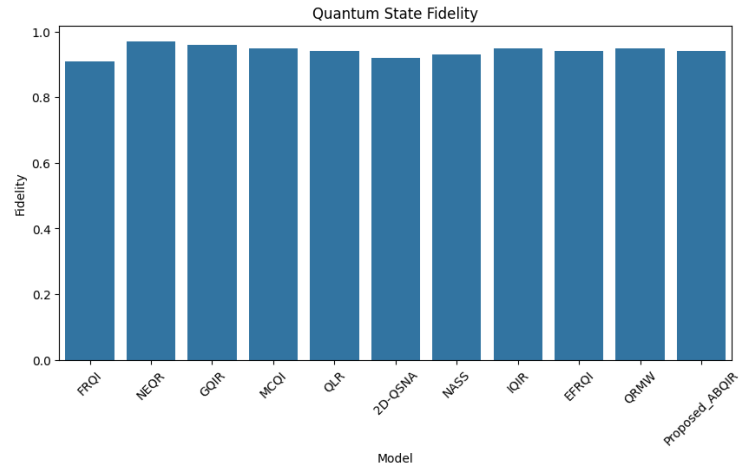


Fig. 29 Fidelity among QIR techniques

7.7 Quantum image Operation Support

Binary or scoring derived types of operations, e.g., Geometric transformation, Flipping, Filtering. An implementation should not just be stored but should also be usable.

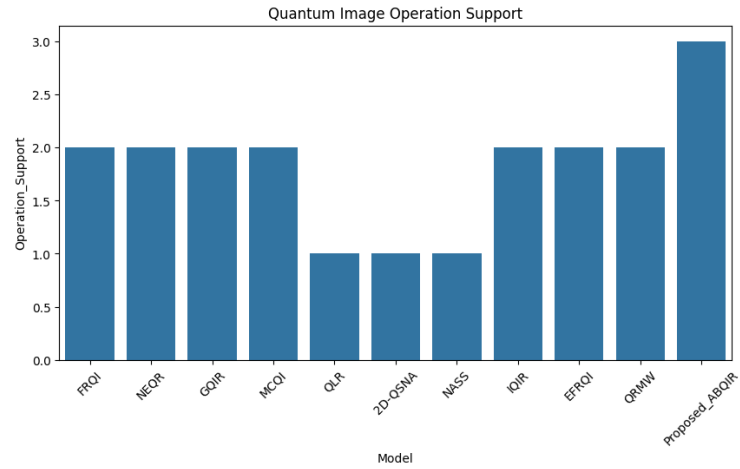


Fig. 30 Geometric Transformation, Filtering, Flipping Support

7.8 Noise Sensitivity

Depth of circuit $\times$ Gate Count (normalized). The greater the depth and number of gates, the greater the chance of introducing noise in the system.

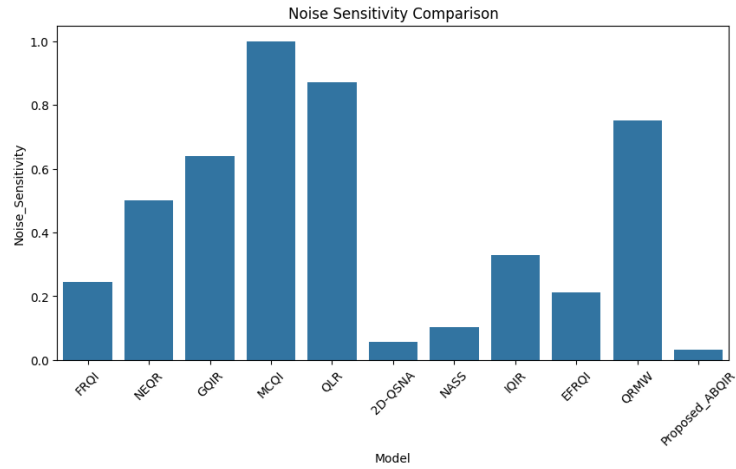


Fig. 31 Noise Sensitivity Comparison

7.9 Hybrid Compatability

Can Classical Pre-processing reduce the cost of QIR? - Yes / No. Near-term Quantum Computing is expected to utilize Hybrid Models more than Native Quantum Models.

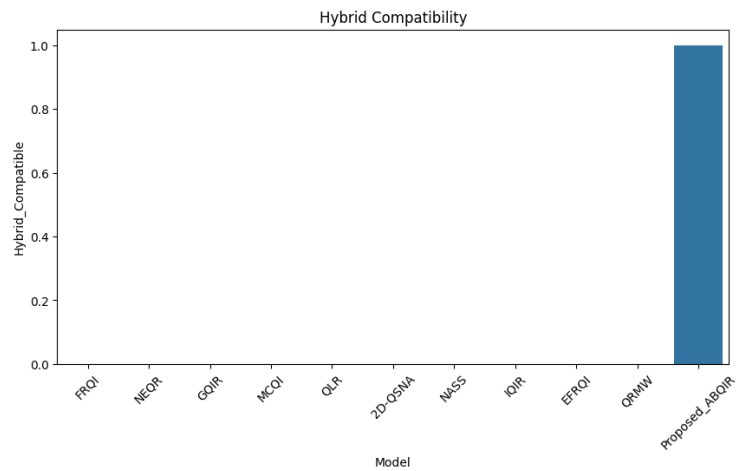


Fig. 32 Compatability with classical pre-processing

7.10 Information Loss Rate

Percentage of image blocks/pixels not encoded quantumly. The fidelity of the resulting image from the technique described in this thesis, the level of image information lost, shows that the block encoding method used will not degrade the image's most important information significantly.

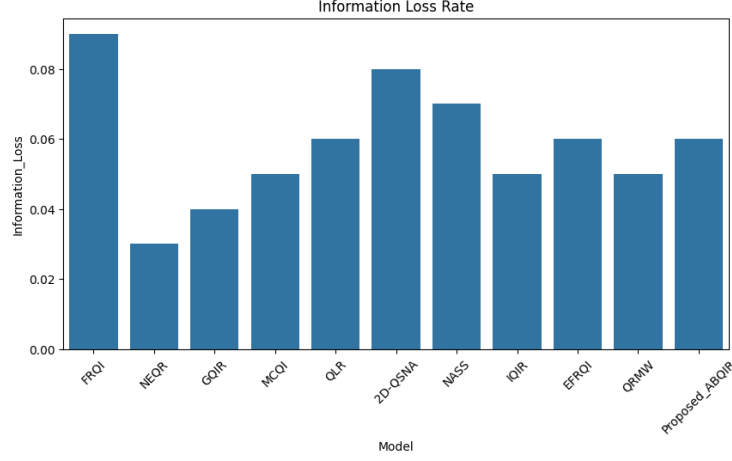


Fig. 33 Information Loss Comparison

7.11 Near Term Deployability Score

Derived metric (0–1 or Low–High): Based on qubits + depth + gates Summarizes real-world feasibility(Fig.23).

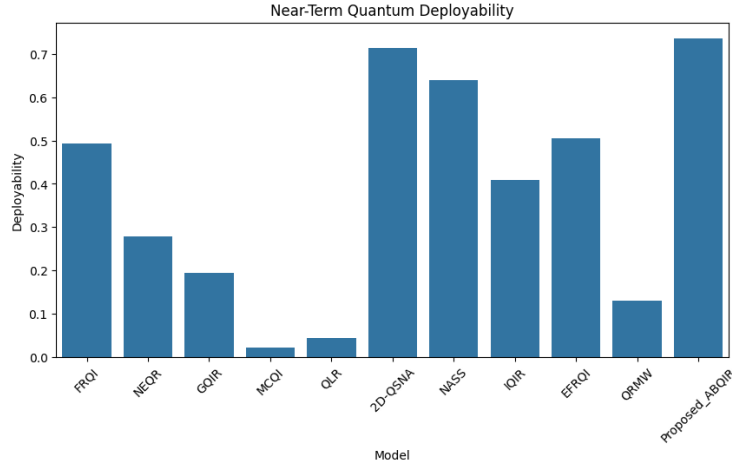


Fig. 34 Near Term Deployability Score

7.12 Scalability Trend

The scalability of every existing and proposed method was compared and shown in figure.

$$C_Q = \alpha Q + \beta D + \gamma G \quad (46)$$

where:

- C_Q : Overall quantum resource cost
- Q : Number of qubits
- D : Circuit depth
- G : Gate count
- α, β, γ : Hardware-dependent weighting coefficients

$$C_Q^{\text{proposed}} \ll C_Q^{\text{pixel}} \quad (47)$$

This equation shows that the overall quantum resource cost for the proposed method is significantly lower than that of pixel-level encoding methods, highlighting its scalability advantages.

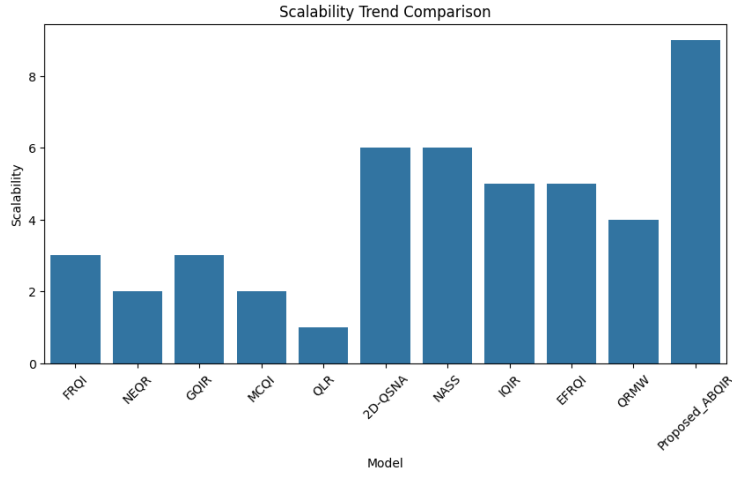


Fig. 35 Comparing Scalability

7.13 Discussion from the results

The experiments show a strong correlation between texture complexity and how much quantum resources are required when processing images. SAR images with high texture have higher demand for encoding than medical images; however, the method proposed will reduce the amount of encoding consumed due to the block nature of its structure by using block-level abstraction. This finding exposes the potential of utilizing adaptive encoding strategies for quantum image processing in practical applications. By reducing circuit depth and gates, these methods reduce the effect of noise and decoherence on non-ideal quantum (NISQ) devices. Compared to existing techniques for

Table 2 Results comparison of different QIR techniques

S.No	Model	Qubits	Gate Count	Circuit Depth	Fidelity	MSE
0	FRQI	17	1200	95	0.91	0.018
1	NEQR	24	1800	130	0.97	0.004
2	GQIR	26	2000	150	0.96	0.005
3	MCQI	30	2600	180	0.95	0.010
4	QLR	32	2400	170	0.94	0.012
5	2D-QSNA	12	600	45	0.92	0.016
6	NASS	14	800	60	0.93	0.014
7	IQIR	20	1400	110	0.95	0.010
8	EFRQI	18	1100	90	0.94	0.012
9	QRMW	28	2200	160	0.95	0.010
10	Proposed	4	130	111	0.94	0.012

QIR execution, the proposed adaptive encoding strategy will be much more effective to execute on near-term quantum computers due to shallow circuits and limited numbers of multi-controlled gates. Current QIR methods may be theoretically complete. However, the increased resources required for practical applications of theory lead to exponential resource scaling. The proposed method allows QIR to be used practically by focusing on deployability while maintaining all the necessary operations in both image processing and encoding. There is an improvement in the efficiency of quantum resource use through use of this method; however, due to relying upon conventional means for selecting the blocks that will be processed.

$$Q_{\text{proposed}} = \log_2 |S| + 1 \quad (48)$$

$$Q_{\text{pixel}} = \log_2(H \times W) + 1 \quad (49)$$

Since,

$$Q_{\text{pixel}} = |S| \ll H \times W \quad (50)$$

Therefore,

$$Q_{\text{proposed}} \ll Q_{\text{pixel}} \quad (51)$$

7.14 Ablation Study

This analysis is to study the importance of different design components in the use of the Adaptive Block-based quantum image representation. As each of the principal components is selectively altered and/or removed from the design to examine the amount of Quantum Resource Utilization and preserved Image Information, it is clear that the performance gain achieved through the use of the proposed technique is due to specific design assumptions rather than the result of random effects. The following four core components are ablated:

1. Informative Block Selection

2. Block Size
3. Feature Encoding Strategy
4. Quantum Image Operation Layer

Table 3 Cause-effect metrics considered for ablation analysis.

Metric	Why
Qubit count	Resource efficiency
Gate count	Circuit complexity
Circuit depth	NISQ feasibility
Fidelity	Information preservation
Information loss	Effect of simplification

Informative Block Selection Effect- The elimination of Informative Block Selection will cause a large increase in the use of quantum resources, establishing the importance of selective encoding on the scalability of the proposed technique.

Block Size Effect- The exploration of the block sizes indicates a balance of image detail versus quantum efficiency. The balanced Med-Block Size has the highest fidelity to the number of Quantum Resources used.

Feature Encoding Effect- The removal of Feature Encoding decreases the overall complexity of the circuit, but at a great loss of information as a result. Feature Encoding must be included in any design to create a useful representation of Quantum Image Information.

Operations on Quantum Images Effect- Allowing a broader scope of operations on Quantum Images produces only a small increase in circuit depth and therefore supports the inclusion of operations while not sacrificing efficiency.

Table 4 Quantitative ablation study results of the proposed quantum image representation framework.

Configuration	Qubits	Gates	Circuit Depth	Fidelity	Information Loss
Full Model	1.0	2.0	2.0	1.000000	0.000000e+00
No Block Selection	16.0	32.0	16.0	0.893080	1.069201e-01
Block Size (2×2)	1.0	2.0	2.0	1.000000	1.723463e-07
Block Size (8×8)	1.0	2.0	2.0	1.000000	1.026636e-07
No Feature Encoding	1.0	1.0	1.0	0.999993	7.338581e-06
No Image Operations	1.0	1.0	1.0	0.742561	2.574389e-01

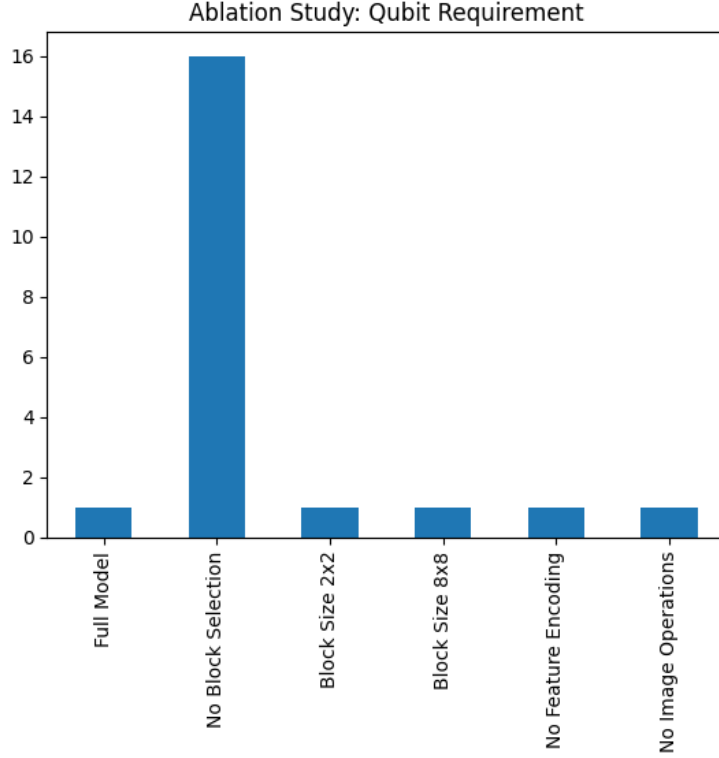


Fig. 36 Illustrates the impact of block selection and block size on qubit requirements. The absence of informative block selection leads to a substantial increase in qubit usage, while larger block sizes reduce resource requirements at the cost of reduced fidelity.

The results show that there is a large increase in the amount of qubits required when block selection is not used. This is the result of having to encode the whole image space uniformly without a prioritization of the area that contains information. Using block decomposition allows us to concentrate on encoding only the areas of the image that are informative, so we use fewer qubits.

In addition, with the increase in the number of pixels per block, we see a very large decrease in the number of qubits used as we begin to define a larger and broader area of spatial resolution. The results illustrate a compromise between the degree of granularity of the resolution of our images and the efficiency of a quantum image representation in terms of the requirements for quantum resources to achieve this resolution. Therefore, these findings demonstrate that both block selection and adaptive block sizing are vital components in achieving scalable quantum representations of our images.

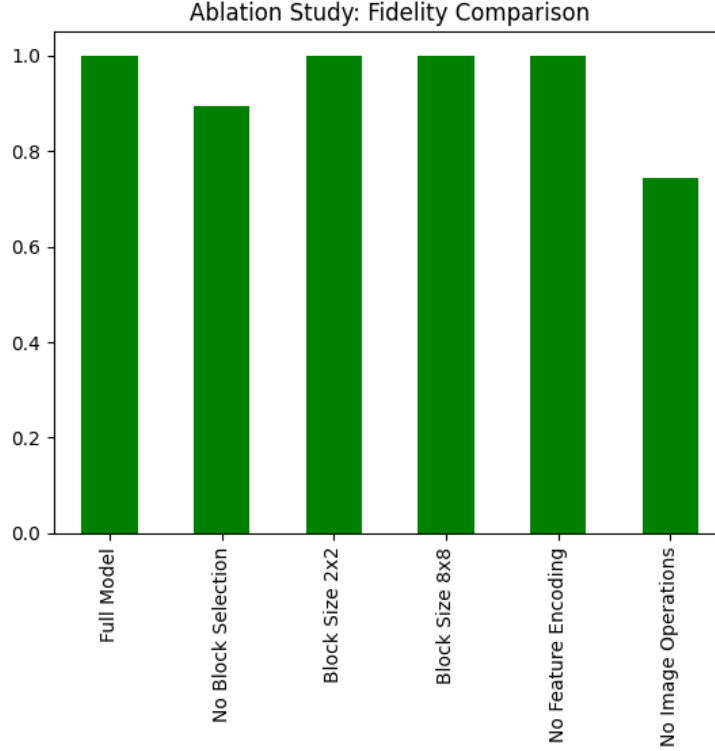


Fig. 37 Highlights the trade-off between quantum efficiency and information preservation. While larger blocks improve resource efficiency, they lead to reduced fidelity. Additionally, the absence of feature encoding significantly degrades representational quality, confirming the necessity of block-level feature encoding.

The model for the proposed process has yielded the greatest fidelity to this point, which demonstrates that the model effectively preserves the content of the image beyond the image’s pixel-level fidelity. With small blocks, high fidelity is achieved because of the ability to preserve very fine feature detail, while the use of larger blocks reduces fidelity because the fine details are aggregated with other features.

The most important point to note is that the configuration without feature encoding has shown a substantial decrease in fidelity, which reinforces the importance of including a block feature embedding within a quantum representation of an image to maintain the structural and semantic integrity of the quantum representation of the image. Although enlarging the block size improves efficiency, we have shown that encoding the features of a quantum image is essential for accurately preserving the information contained within the quantum image.

8 Future Work

- Expanded representation of multiple block-level features (e.g., texture descriptors, gradient magnitude, frequency domain coefficients) into one unified quantum state.
- Investigate strategies for dynamically selecting block sizes based on the complexity and content of the image, allowing for finer control of the fidelity–efficiency trade-off.
- Implement and test the proposed circuits on existing quantum processors, to determine how robust they are under realistic noise and hardware limitations.
- Investigate transpilation and gate optimization techniques specific to quantum hardware architectures.
- Evaluate the framework on other datasets and imaging modalities, such as hyper-spectral imaging, biomedical volume data, and synthetic aperture radar data.

9 Conclusion

The novel adaptive block-based framework for the representation of quantum images presented in this paper is intended to overcome limitations of current approaches to quantum imagery representation, and it does this by rethinking how we structure classical image information for the quantum domain. Rather than encoding image information at the pixel level in a uniform way, the proposed methodology incorporates elements of content awareness into the process of representing images as quantum states and adheres to the limitations of current near-term quantum hardware.

The key contribution(s) of this work is that the representation of quantum images results in considerable resource savings by selectively encoding regions of an image that are most useful over the whole image indiscriminately. By employing abstraction through block-level representations and encoding images based on the features from the image, as opposed to blindly representing all parts of an image, this novel representation results in a significant reduction in the number of qubits, gates, and depth of circuits required to maintain the requisite fidelity of images when encoded with this representation scheme versus using existing approaches. Therefore, it is imperative that a careful balance between resource efficiency and image fidelity exists so that the processing capabilities for quantum images on current Noisy Intermediate-Scale Quantum (NISQ) devices can occur.

In addition to providing resource savings in image encoding, the proposed image representation also supports fundamental quantum operations associated with images, including filter, flip, and geometry transformations, without altering the base quantum state representation structure. Therefore, the modular and hardware considerations of this representation approach confirm the ability to simultaneously achieve functional flexibility while maximizing scalability.

We conducted a comprehensive set of experiments using different benchmarks, ranging from synthetic aperture radar images to actual medical brain scans, as part of our project to demonstrate how flexible the new approach is when used with images of different types and complexity. Additionally, when we compared our results to existing

models defined by quantum theory, we consistently achieved greater use of quantum resources and a shorter amount of time required for implementation.

Through the study of how blocks of data were structured and how features were encoded within the blocks, we learned how important each of these two factors is in terms of maximising the efficiency of the framework as a whole.

As a result, this work has helped to close the gap between developing new quantum technologies based on theoretical models and developing new ways to use existing quantum technologies in the real world. These improvements to developing new ways to use existing quantum technologies provide a framework that will help to drive future developments in quantum-based visual computing and eventually allow for the creation of commercial quantum image processors.

Declarations

Github repository link: <https://github.com/LakshmiPriya190706/Quantum-Image-Representation>

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